Analysis of Modulation Schemes Applied to the Electrical and Optical Systems of an LED Load

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***Abstract*—In this work, an approach using transfer functions that describe the electrical and optical behavior of an LED load was proposed. A viable alternative was presented for studying modulation schemes on a real LED load, aiming to explore ways to emulate scenarios closer to the reality of VLC applications. Three modulation schemes were used, the energies of the optical signals were calculated, and it was concluded that up to 1 MHz, PWM modulation contains more energy compared to 4-PAM modulation with the same maximum amplitude.**

***Index Terms*—LED, electrical dynamics, optical dynamics, modulation schemes, signal energy**

I. Introduction

Over the past few years, the lighting industry has seen significant development in luminaire technologies with solid-state lighting (SSL). Light-emitting diodes (LEDs) have come to dominate the lighting market and are now present in more than 50% of households globally [1]. One of the main reasons for this is their advantages over other lighting sources, among which stand out their low energy consumption, high efficacy, and long lifespan [2].

The continuous improvement and the steady increase in the use of LEDs make the employment of this technology favorable beyond its primary purpose, which is to illuminate the environment, thus opening space for other applications. For example, Visible Light Communication (VLC) takes advantage of the existing infrastructure for data transmission based on the intentional modulation of light. This occurs because LEDs have the capability to respond to high-frequency electrical stimuli, allowing the transfer of the content of the electric current to the generated light [3]–[5].

General-purpose lighting with SSL is predominantly composed of phosphor-covered LEDs (PC-LED) [6], but technologies such as color-mixed LEDs (CM-LED) or RGB (*Red, Green, Blue*) are promising in terms of luminous efficacy and data transmission rate [7]–[9]. VLC is a high-speed, short-range wireless communication solution, but its bandwidth is limited, particularly by the light source. PC-LEDs have a bandwidth of approximately 3 MHz [10]–[12] due to the slow response of the phosphor layer. In contrast, RGB LEDs, as they do not have a phosphor coating, offer a unique advantage: depending on the specific LED, they can exhibit a bandwidth more than 5 times greater compared to the same LED with phosphor coating [13].

Besides the phosphor issue, other electrical and optical characteristics, such as size and power density, affect the response by defining factors like electrical parasitic elements, operating temperature, and nominal current. It is known that the higher the system bandwidth, the higher the achievable data rate. In this regard, understanding the dynamics of the light emitter is essential to design an effective VLC transmitter. This can only be accomplished with suitable models for accurate simulations, considering the real system behavior. However, manufacturers do not provide the dynamic (AC) model of the LED, since these devices have not yet been produced for communication applications [14].

Regarding the physical layer of a VLC system, the light source is one of its main components. This is due to the fact that, besides being the transmitter of the modulated signal, in most applications — especially those using PC-LEDs — it is the limiting factor of the communication system's bandwidth, owing to its slower response compared to the other system elements [12].

To design high-performance circuits, the better the understanding of the dynamics of the components involved, the more accurate the calculations and simulations will be compared to the results obtained in practice. Thus, high-frequency LED models are essential for the implementation of advanced VLC technologies, since by understanding the high-frequency dynamics of LEDs, it is possible to improve the performance and reliability of these technologies in various applications.

When a VLC system is integrated into a luminaire, energy consumption increases. In this regard, the study of energy efficiency in a VLC system is of significant importance [15]. The analysis of modulation techniques, coupling systems, amplification of the modulated signal, and the choice of light sources are some of the approaches that can be investigated with the aim of developing more efficient VLC systems.

In [14], the author characterizes the electrical and optical dynamics of an LED load. One of the presented results is the transfer functions representing the electrical and optical dynamic systems of the load. Thus, the objective of this work is to analyze the energy of a signal after passing through the electrical and optical systems, comparing the energy of the same signal sent with different modulation techniques, ultimately comparing which technique will present the highest energy value, since the higher the energy of a signal, the easier it will be to identify it at the receiver. Therefore, this work aims to analyze the feasibility of using the electrical and optical systems proposed in [14] in the comparative study between different modulation schemes.

This work is organized as follows: Section II presents the methodology proposed for the development of this research; Section III discusses the algorithm used; Section IV exposes the results obtained; finally, Section V presents the main conclusions regarding the work.

II. Signal Analysis Methodology

In [14], the author conducts the characterization of the electrical and optical behavior of an LED load. This load was characterized on a test bench, and the collected data were processed using mathematical software. Table I presents the main considerations regarding the load and the characterization conditions. At the end of the process, the transfer functions corresponding to the electrical and optical response of the LED load were obtained.

TABLE I
LED System Parameters [14]

Parameter	Definition
LUXEON 3014 LED	3 strings / 16 LEDs
Z_{DC}	24.6 Ω
DC Bias	200 mA and 49 V → 9.8 W
Frequency Range	10 kHz – 10 MHz
Superimposed Signal (AC)	$I_{\mathrm{Peak\,AC}} = 50$ mA

The electrical dynamics correspond to an impedance and describe the behavior of the LED current for a given voltage signal applied to it. Equation (1) is the transfer function of the electrical dynamics, restructured from the data in [14].

$$\frac{V(s)}{I(s)} = \frac{1212s^2 + 8.064 \times 10^{10}s + 2.597 \times 10^{15}}{s^2 + 2.615 \times 10^9 s + 8.624 \times 10^{13}} \quad (1)$$

In turn, the optical dynamics represent the amount of light generated by the load that is received by the receiver and converted into voltage at its output. The produced light is directly proportional to the current flowing through the LED; thus, the optical dynamics directly relate the LED current to the output voltage of the photodetector. Equation (2) is the transfer function of the optical dynamics, restructured from the data in [14]. It is worth noting that the author mentions it is a phosphor-coated LED and the distance between the emitter and the receiver was 4 *cm*.

$$\frac{V(s)}{I(s)} = \frac{-0.0301\, s^2 + 3.006 \times 10^6\, s + 4.185 \times 10^{12}}{s^2 + 1.412 \times 10^7\, s + 1.169 \times 10^{13}} \quad (2)$$

This work proposes a method for analyzing different modulations for a specific LED load. As previously mentioned, depending on the modulation used, it may imply different efficiencies for the same VLC application scenario. In this regard, aiming to verify the signal energy of different modulations on a load more faithful to reality, the electrical (1) and optical (2) dynamics are used jointly.

The analysis is performed entirely in mathematical software; in this work, MATLAB was chosen. Initially, a voltage signal V_{mod} is generated, representing the modulated signal to be transmitted. This signal is applied at the input of the electrical dynamics system; since it is a voltage signal, the transfer function (1) must be inverted so that, when the voltage signal and the system's transfer function are multiplied in the frequency domain, the resulting signal is a current signal I_{AC}.

In turn, the current signal resulting from (1) represents the AC current passing through the LED as a function of the modulated voltage signal. This signal is applied at the input of the optical dynamics system. The output resulting from passing the current signal I_{AC} through this system will be a voltage signal V_{opt}. This signal corresponds to the voltage generated by the photodetector due to the light produced by the modulated current passing through the LED. Fig. 1 summarizes the steps described above.

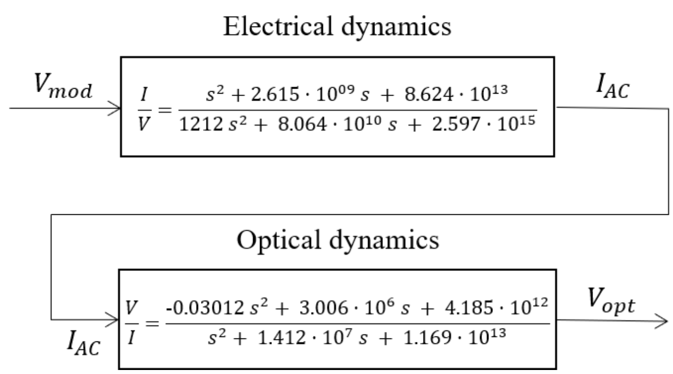

Fig. 1. Signal processing flow.

The comparison criterion consists of analyzing the energy of each voltage signal V_{opt}, originating from each type of modulation used. The energy can be calculated by equation (3).

$$E_g = \int_{-\infty}^{\infty} g^2(t)\, dt \quad (3)$$

III. Algorithm Structure

The algorithm for executing the signal processing flow described in Fig. 1 begins with the definition of the sampling frequency, which was set to 1 GHz, and with the definition of the transfer functions, which are discretized in a subsequent step using the Tustin method. Figures 2 and 3 present, respectively, the Bode diagrams of the discretized electrical and optical dynamic systems.

Next, the symbols to be transmitted are configured; a total of 1,000 symbols was set, each composed of 2 bits, and the modulation signal frequency is defined as desired.

The modulations selected for analysis were:

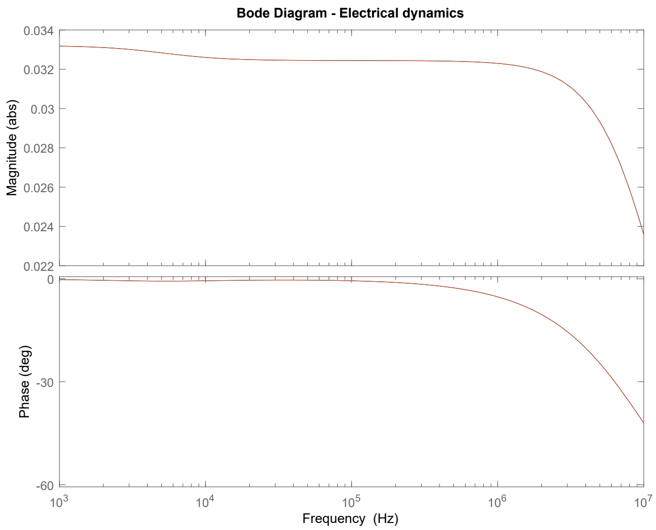

Fig. 2. Bode diagram of the electrical dynamics.

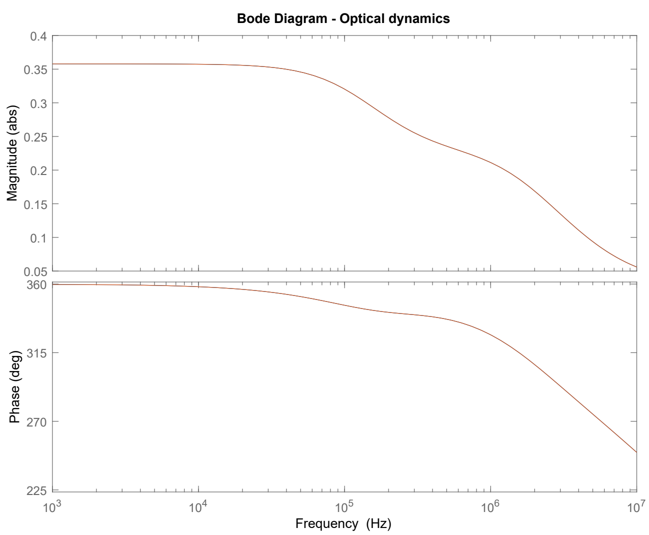

Fig. 3. Bode diagram of the optical dynamics.

- **4-PAM from 0 V to 3 V:** symbols are equally divided between 0 V and 3 V. The choice of 4-PAM modulation is due to the energy efficiency that M-PAM modulation demonstrated in [15]. The symbol mapping corresponds to:
  - 00 $\rightarrow$ 0
  - 01 $\rightarrow$ 1
  - 10 $\rightarrow$ 2
  - 11 $\rightarrow$ 3

Figure 4 shows a sequence of 10 symbols using 4-PAM modulation from 0 V to 3 V at 1 MHz.

- **4-PAM from 0 V to 1 V:** symbols are equally divided between 0 V and 1 V; this modulation was chosen to verify, due to its lower amplitude, the expected lower energy

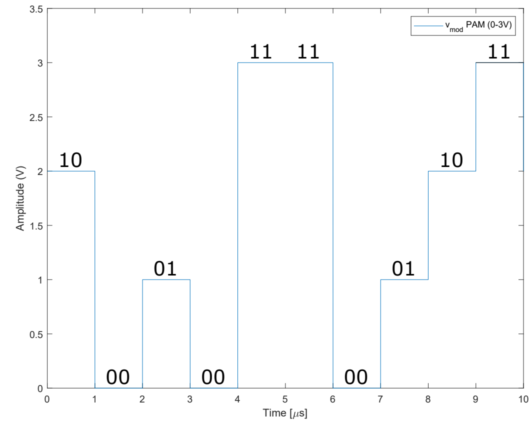

Fig. 4. Sequence of 10 symbols using 4-PAM (0–3 V).

value compared to the previous similar modulation. The symbol mapping corresponds to:
  - 00 $\rightarrow$ 0
  - 01 $\rightarrow$ 0.33
  - 10 $\rightarrow$ 0.66
  - 11 $\rightarrow$ 1

Figure 5 shows a sequence of 10 symbols using 4-PAM modulation from 0 V to 1 V at 1 MHz.

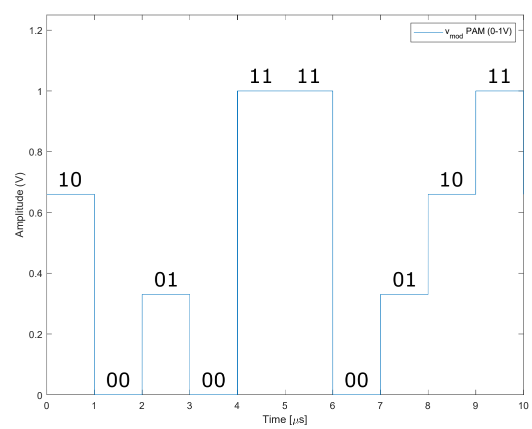

Fig. 5. Sequence of 10 symbols using 4-PAM (0–1 V).

- **PWM:** the signal excursion ranges from 0 V to 1 V, with each symbol indicated by a specific pulse width. The symbol mapping corresponds to:
  - 00 $\rightarrow$ 20% pulse width
  - 01 $\rightarrow$ 40% pulse width
  - 10 $\rightarrow$ 60% pulse width
  - 11 $\rightarrow$ 80% pulse width

Figure 6 shows a sequence of 10 symbols using PWM modulation at 1 MHz.

The MATLAB function *lsim* provides the response of a continuous or discrete linear system to an arbitrary time-domain input, where the response is also a time-domain signal.

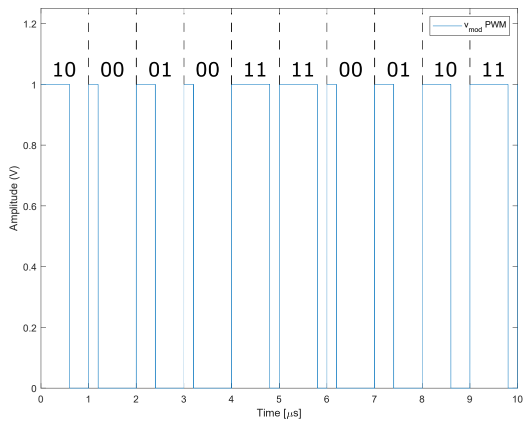

Fig. 6. Sequence of 10 symbols using PWM.

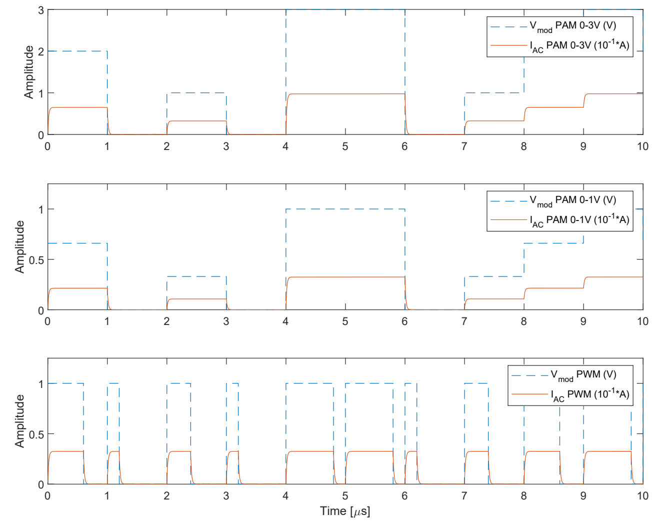

Fig. 7. Effect of the electrical system on the input signal. Input: V_{mod} (Blue); Output: I_{AC} (Orange).

This function is used to obtain the responses of the electrical and optical dynamics, using a voltage signal as input for the former and a current signal for the latter. Finally, the energy of each signal is calculated.

IV. Results

The algorithm was executed for three different frequency values (500 kHz, 1 MHz, and 5 MHz), since, as shown in Fig. 2 and Fig. 3, the system gain decreases with increasing frequency. In this context, it becomes relevant to analyze the behavior of different modulations at distinct frequencies.

Waveforms will be presented only for one acquisition at 1 MHz, aiming to demonstrate the behavior of a signal as it passes through the electrical and optical systems. The input signals to the electrical system and their respective outputs for each modulation scheme are shown in Fig. 7. Similarly, the input and output signals of the optical system are presented in Fig. 8.

It is observed that the electrical system, for a 1 MHz signal, exhibits electrical dynamics with minimal delay, since the current signal is minimally distorted. In contrast, the optical system behaves more slowly, as expected, because, as previously mentioned, the phosphor-coated LED has a slow dynamic response.

To cover all proposed tests, the algorithm was executed 9 times, with three acquisitions for each frequency, and the average value was calculated at the end. The obtained values are presented in Tables II and III.

At lower frequencies, due to reduced attenuation, the signals exhibit higher energy. Among the presented modulation schemes, the 4-PAM (0–3 V) has the highest energy, which was expected since the input signal energy is already greater, as observed by the higher amplitude. Among the schemes with amplitudes ranging from 0 to 1 V, PWM showed higher energy at 500 kHz and 1 MHz; however, at 5 MHz, the 4-PAM (0–1 V) modulation exhibited more energy.

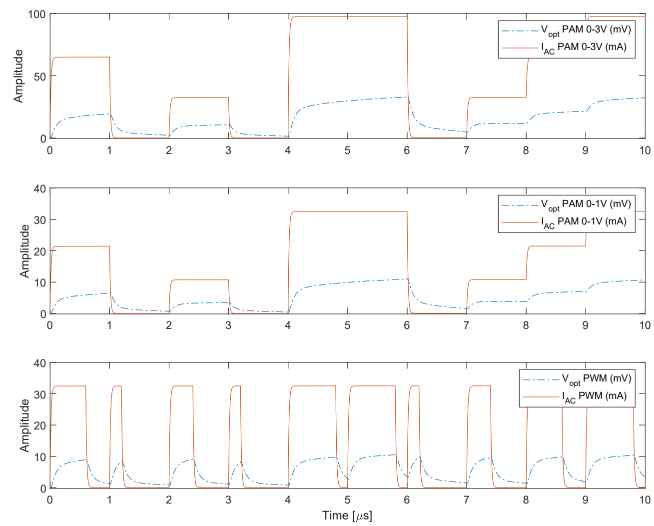

Fig. 8. Effect of the optical system on the input signal. Input: I_{AC} (Orange); Output: V_{opt} (Blue).

V. Conclusions

This work conducted an analysis of the energy of different modulations on an LED load, represented by its transfer functions that correspond to the electrical and optical dynamics of the load. Through the proposed algorithm, the correct relationship between the input signals and the systems was observed, as the gains and phase shifts align with expectations.

Based on the energy analysis, it can be stated that at lower frequencies there will be systems with higher energy, but not necessarily more efficient. Among the modulations, considering the same maximum signal value, PWM exhibits higher energy up to 1 MHz.

Based on the obtained data, the use of the transfer functions

TABLE II
ACQUIRED VALUES FOR EACH MODULATION SCHEME

Frequency	Modulation	Energy (J)		
		1st	**2nd**	**3rd**
500 kHz	PAM 0–3 V	8.628E-07	8.818E-07	8.971E-07
	PAM 0–1 V	9.513E-08	9.721E-08	9.891E-08
	PWM	9.957E-08	1.009E-07	1.022E-07
1 MHz	PAM 0–3 V	4.168E-07	4.075E-07	4.376E-07
	PAM 0–1 V	4.596E-08	4.496E-08	4.821E-08
	PWM	4.663E-08	4.573E-08	4.837E-08
5 MHz	PAM 0–3 V	7.081E-08	7.079E-08	7.240E-08
	PAM 0–1 V	7.798E-09	7.796E-09	7.975E-09
	PWM	7.555E-09	7.543E-09	7.663E-09

TABLE III
AVERAGE OF THE ACQUIRED VALUES FOR EACH MODULATION SCHEME

Frequency	Modulation	Energy (J)
		Média
500 kHz	PAM 0–3 V	8.805E-07
	PAM 0–1 V	9.708E-08
	PWM	1.009E-07
1 MHz	PAM 0–3 V	4.206E-07
	PAM 0–1 V	4.638E-08
	PWM	4.691E-08
5 MHz	PAM 0–3 V	7.133E-08
	PAM 0–1 V	7.856E-09
	PWM	7.587E-09

presented in [14] is validated for application in analyses of different AC signal scenarios on the characterized LED load. This enables work on the electrical and optical contexts either separately or jointly.

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